

The empirical recurrence for OEIS A297462: recovering a conjecture that is not written down

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Abstract

OEIS A297462 records “Empirical recurrence of order 72”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 276$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 72, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 72 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297462 is “Number of $6 \times n$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 0 or 2 neighboring 1s.”. It has offset 1 and begins

64, 299, 6888, 55713, 862112, 8800153, 116817805, 1310421193, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Dec 30 2017 11:13:56, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 276$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 276$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 552$ exact terms of the model returns order 72 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	11	c_{19}	684476215027	c_{37}	−702279038054852	c_{55}	470974229957739
c_2	107	c_{20}	1438964442015	c_{38}	−5975489952241530	c_{56}	350943736214504
c_3	−1097	c_{21}	2706768259547	c_{39}	−7163906172604830	c_{57}	258667511273183
c_4	−2798	c_{22}	1918637981153	c_{40}	−9729902509670301	c_{58}	−278249378530947
c_5	26574	c_{23}	−18967411235777	c_{41}	425687986553823	c_{59}	−221926904044591
c_6	−8777	c_{24}	−37545607808191	c_{42}	9751770868679593	c_{60}	44164771238865
c_7	196042	c_{25}	−34618041323539	c_{43}	5834828853234531	c_{61}	61612280791352
c_8	647393	c_{26}	3002374246692	c_{44}	5536181007905654	c_{62}	28100686806256
c_9	−9836817	c_{27}	254955659851602	c_{45}	376557710984838	c_{63}	−8095265732053
c_{10}	995400	c_{28}	444795647652035	c_{46}	−8270108304550693	c_{64}	−10513189639898
c_{11}	−9887716	c_{29}	196440134722447	c_{47}	−3867144231050355	c_{65}	−1223817893025
c_{12}	−198338589	c_{30}	−298304282010810	c_{48}	950625490439796	c_{66}	731882201951
c_{13}	1923491078	c_{31}	−1729783389549665	c_{49}	979106391826694	c_{67}	836801272264
c_{14}	3340426284	c_{32}	−2409723470697175	c_{50}	2613727643227950	c_{68}	3647092213
c_{15}	−8654146529	c_{33}	−116793399553783	c_{51}	1433291492098234	c_{69}	−55111484696
c_{16}	−16997228353	c_{34}	2001669382358730	c_{52}	−1838686329746826	c_{70}	−12978286164
c_{17}	−109792950580	c_{35}	5125041383592210	c_{53}	−1581125314416109	c_{71}	561120
c_{18}	−143624742858	c_{36}	6770173376435993	c_{54}	325566220109203	c_{72}	147492800

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 72, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $72 + 4$ terms removed returns order 72 as well, so $\deg q_{\min} = 72 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $72 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 72 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A297462.
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